
Knowledge creation: a meta-framework

Nilmini Wickramasinghe

Stuart Graduate School of Business, Illinois Institute of Technology,
565 W Adams St, Suite 406, Chicago, IL 60661, USA
Fax: +1 312.906.6549
E-mail: nilmini@stuart.iit.edu

Abstract: In today's knowledge based economy, sustainable strategic advantages are gained more from an organisation's knowledge assets than from the more traditional types of economic resources (Drucker, 1993). The paper discusses the multi-faceted nature of the knowledge construct including its subjective and objective as well as tacit and explicit components. Moreover, technical theories pertaining to knowledge creation such as data mining are brought together with social or people oriented theories on knowledge creation including Nonaka's knowledge spiral to develop a meta-framework in which to view key processes, tools and techniques that facilitate effective knowledge creation and its management.

Keywords: knowledge management; data mining; knowledge spiral; knowledge discovery; meta-framework; tacit knowledge; explicit knowledge.

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Biographical notes: Nilmini Wickramasinghe researches and teaches in several areas within Information Systems including knowledge management, e-commerce and m-commerce and organisational impacts of technology. In addition, She focuses on the impacts of technologies on the healthcare industry. She is well published in all these areas and as well as numerous journal papers has co-edited one book *Creating Knowledge-based Healthcare Organisations* published by Idea as well as having contributed several book chapters and encyclopedia entries. She regularly presents her work throughout North America as well as in Europe and Australasia. She is the US representative of the Health Care Technology Management Association (HCTM), an international organisation that focuses on critical healthcare issues and the role of technology within the domain of healthcare, and holds an Associate Professor position at the Stuart Graduate School of Business, Illinois Institute of Technology.

1 Background

An extensive review of the knowledge management literature (Schultze and Leidner, 2002; Massey et al., 2002, p.271) confirms that '... there has been very little research on how to successfully develop and implement KM solutions ...'. Furthermore, from these studies, it appears that currently there is little, if any, research on how to actualise the knowledge spiral and thereby facilitate the creation of new knowledge, a

significant component within any successful knowledge management solution. In addition, a study by Alavi and Leidner (2001, p.117) suggests “that considering the flexibility of modern IT, other forms ... of knowledge creation can be enhanced through the use of various forms of information systems”. The collective message from these findings indicates the need for a more rigorous use of information technologies and techniques in facilitating successful knowledge management initiatives, especially knowledge creation/acquisition. This paper aims to address some of these unresolved issues and challenges concerning knowledge creation/acquisition, by developing a meta-framework that combines the major knowledge creation/acquisition processes namely, Nonaka’s people oriented knowledge spiral with the technology driven data mining approach to knowledge evolution (Yen et al., 2004). In so doing, a unified holistic approach for structuring knowledge creation/acquisition is presented.

Specifically, the paper demonstrates how data mining not only can reveal knowledge hidden in large amounts of data assets, but also can effect the four key knowledge transformations as identified by Nonaka (1994):

- transforming existing explicit knowledge into new explicit knowledge
- transforming existing explicit knowledge into new tacit knowledge
- transforming existing tacit knowledge into new explicit knowledge
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These transformations constitute the knowledge spiral wherein there is a continuous flow from extant (tacit and/or explicit) knowledge to new knowledge (tacit and/or explicit).

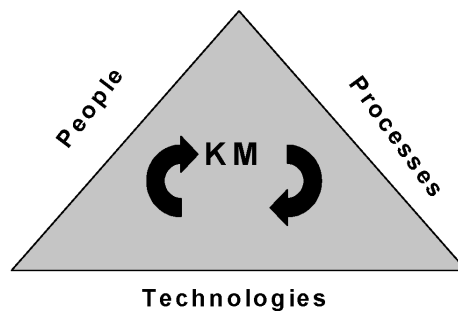
The rest of the paper is organised as follows: Section 2 discusses knowledge management, the types of knowledge and the knowledge spiral. Section 3 discusses data mining concepts and techniques as well as the role of data mining in knowledge creation. Section 4 introduces the meta-framework. Section 5 illustrates the potential benefits of the proposed meta-framework using a sample data set, while Section 6 provides discussions, followed by a conclusion.

2 Knowledge management

Knowledge management is a key approach to help solve current business problems such as competitiveness and the need to innovate that are faced by organisations today (Mentzas, 2004), as well as being considered a competitive necessity (Wickramasinghe and Sharma, 2005). The premise for knowledge management is based on a paradigm shift in the business environment where knowledge is central to organisational performance (Wickramasinghe and Sharma, 2004; Drucker, 1993). This macrolevel paradigm shift also has significant implications upon the microlevel processes of assimilation and implementation of knowledge management concepts and techniques (Swan et al., 1999) i.e., the Knowledge Management Systems (KMS) that are currently in place. In essence, knowledge management not only involves the production of information but also the capture of data at the source, the transmission and analysis of this data as well as the communication of information based on or derived from the data to those who can act on it (Davenport and Prusak, 1998).

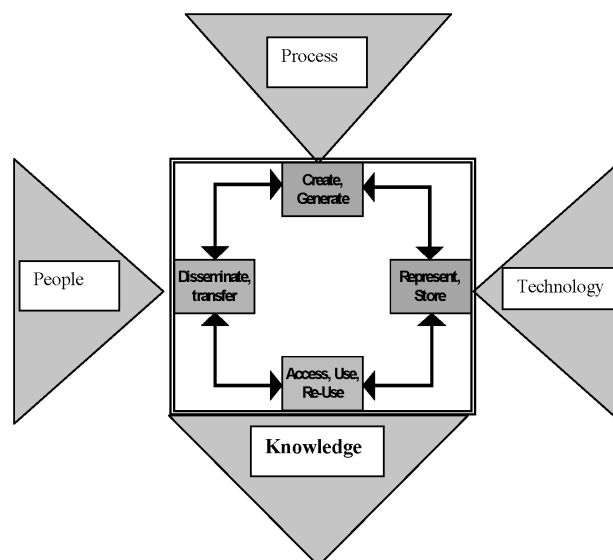
Knowledge management represents a confluence of many bodies of literature including management, organisational behaviour and information systems. Thus successful knowledge management initiatives require a tripartite view; namely the incorporation of people, processes and technologies (Massey et al., 2002). We can visualise this in terms of the KM Triad as shown in Figure 1. The significance of the KM Triad is to emphasise that knowledge can be created by people and/or technologies and can also be embedded in processes.

Figure 1 The KM triad



Broadly speaking, knowledge management involves four key steps of creating/generating knowledge, representing/storing knowledge, accessing/using/reusing knowledge and disseminating/transferring knowledge (Swan et al., 1999; Davenport and Prusak, 1998; Massey et al., 2002). By combining the KM Triad with these four key steps it is possible to form the KM Diamond as shown in Figure 2. The KM Diamond highlights the importance of the impact of the three elements of KM; namely people, process and technology on the four steps of knowledge management. In other words, successful KM initiatives require consideration and interactions among all of these components.

Figure 2 The KM diamond



Knowledge creation, generally accepted as the first step for any knowledge management endeavour (as depicted in Figure 2), requires an understanding of the knowledge construct as well as its people and technology dimensions. A subtle, yet significant aspect of the key process of knowledge management is that the overall process is clearly continuous while the subprocesses or steps are, in fact, discrete in nature. Given that knowledge creation is the first step in any knowledge management initiative, it naturally has a significant impact on the other consequent KM steps (depicted in Figure 2), thus making knowledge creation a key focal point of many frameworks currently in the literature. In order to fully appreciate the need for taking a holistic approach to KM it is important to first discuss the subtleties of the knowledge construct itself.

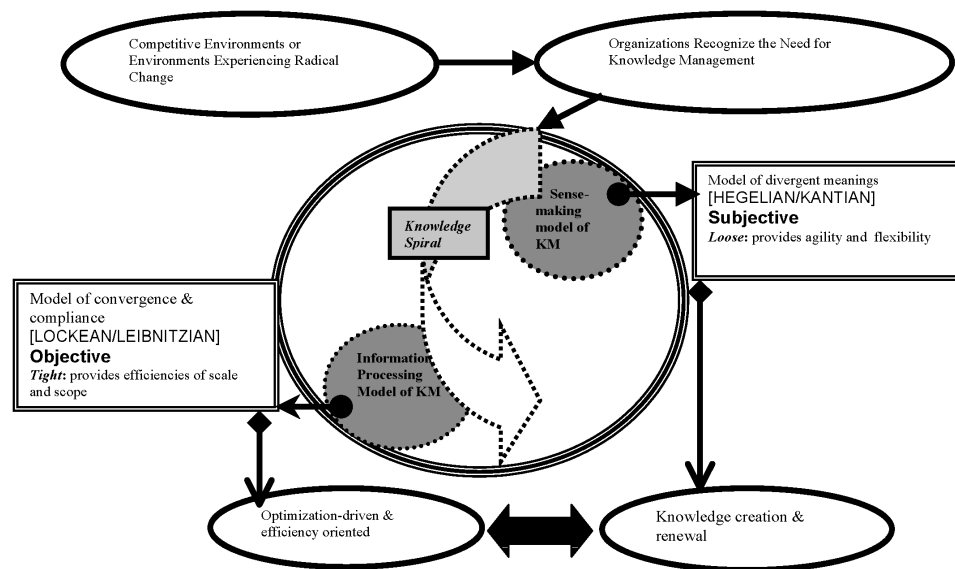
2.1 Types of knowledge

As with many concepts in organisational theory, the existence of duality as discussed by Orlikowski (1992) applies when we examine the types of knowledge and even the processes of knowledge management where we have the four discrete but continuously evolving steps. Traditionally, researchers turn to Burrell and Morgan's (Schultze and Leidner, 2002) well established framework of objective and subjective characterisations. However, a more recent approach elaborated on by Schultze and Leidner (2002) is Deetz's four discourses of organisational inquiry bounded by the dimensions of consensus/dissensus and emergent/a priori. Knowledge, then like so many other concepts in organisational theory, manifests this duality. In trying to manage knowledge, it is necessary to first understand this binary nature of knowledge; namely its objective and subjective components (Alavi and Leidner, 2001) as well as its consensus/dissensus dimensions (Schultze and Leidner, 2002). Specifically, knowledge can exist as an object, in essentially two forms; explicit or factual knowledge and tacit or 'know how' (Paiva, 2003; Polyani, 1958, 1996). It is well established that while both types of knowledge are important, tacit knowledge is more difficult to identify and thus, manage (Nonaka, 1994). Of equal importance, though perhaps less well defined, knowledge also has a subjective component and can be viewed as an ongoing phenomenon, being shaped by social practices of communities (Boland and Tenkasi, 1995). The objective elements of knowledge can be thought of as primarily having an impact on process while the subjective elements typically impact innovation. Enabling and enhancing both effective and efficient processes, as well as the functions of supporting and fostering innovation are key concerns of knowledge management. Thus, we have an interesting duality in knowledge management that some have called a contradiction (Schultze, 1998) and others describe as the *loose-tight* nature of knowledge management (Malhotra, 2000).

The *loose-tight* nature of knowledge management comes to being because of the need to recognise and draw upon two distinct philosophical perspectives; namely, the Lockean/Leibnitzian stream and the Hegelian/Kantian stream. Models of convergence and compliance representing the *tight* side are grounded in a Lockean/Leibnitzian tradition. These models are essential to provide the information processing aspects of knowledge management, most notably by enabling efficiencies of scale and scope and thus supporting the objective view of knowledge management. In contrast, the *loose* side provides agility and flexibility in the tradition of a Hegelian/Kantian perspective. Such models recognise the importance of divergence of meaning which is essential to support the 'sense-making', subjective view of knowledge management. Figure 3 depicts

the Yin-Yang model of knowledge management (Wickramasinghe and Mills, 2001). This figure shows that given a radical change to an environment or given a highly competitive environment (both of which are typical characterisations of today's business environment) an organisation needs knowledge to survive. From the Yin-Yang depiction of knowledge management we see that knowledge is required for the organisation to be effective and efficient but new knowledge and knowledge renewal is also necessary; thereby making both forms of knowledge important for an organisation to capture, in order to truly benefit from knowledge management.

Figure 3 Yin-Yang model of knowledge management



Source: Wickramasinghe and Mills (2001)

2.2 The knowledge spiral

Organisational knowledge is not static; rather it changes and evolves during the lifetime of an organisation. What is more, it is possible to transform one form of knowledge into another; i.e., transform tacit knowledge into explicit and vice versa or transform subjective knowledge into objective knowledge (Wickramasinghe and Mills, 2001). This process of transforming one form of knowledge into another is known as the knowledge spiral (Nonaka, 1994). Naturally, this does not imply one form of knowledge is necessarily transformed 100% into another form of knowledge. According to Nonaka (1994):

- tacit to tacit knowledge transformation usually occurs through apprenticeship type relations where the teacher or master passes on the skill to the apprentice
- explicit to explicit knowledge transformation usually occurs via formal learning of facts

- tacit to explicit knowledge transformation usually occurs when there is an articulation of nuances; for example, if an expert surgeon is questioned as to why he performs a particular surgical procedure in a certain manner, by his articulation of the steps the tacit knowledge becomes explicit
- explicit to tacit knowledge transformation usually occurs when explicit knowledge is internalised and can then be used to broaden, reframe and extend one's tacit knowledge.

Integral to these transformations of knowledge through the knowledge spiral is that new knowledge is being continuously created (Nonaka, 1994) and this can potentially bring many benefits to organisations. In the case of transforming tacit knowledge to explicit knowledge for example, an organisation is able to capture the expertise of particular individuals; hence, this not only expands the organisational memory but also enables single loop and double loop organisational learning to take place (Huber, 1984).

3 Data mining

From a micro perspective, data mining is a vital step in the broader context of the knowledge discovery in databases (KDD) that transforms data into knowledge by identifying valid, novel, potentially useful and ultimately understandable patterns in data (Fayyad et al., 1996). KDD plays an important role in datadriven decision support systems that include query tools, report generators, statistical analysis tools, data warehousing and online analytic processing (OLAP). Data mining algorithms are used on data sets for model building, or for finding patterns and relationships in data. How to manage such newly discovered knowledge, as well as other organisational knowledge assets, is the realm of knowledge management.

3.1 Steps in data mining

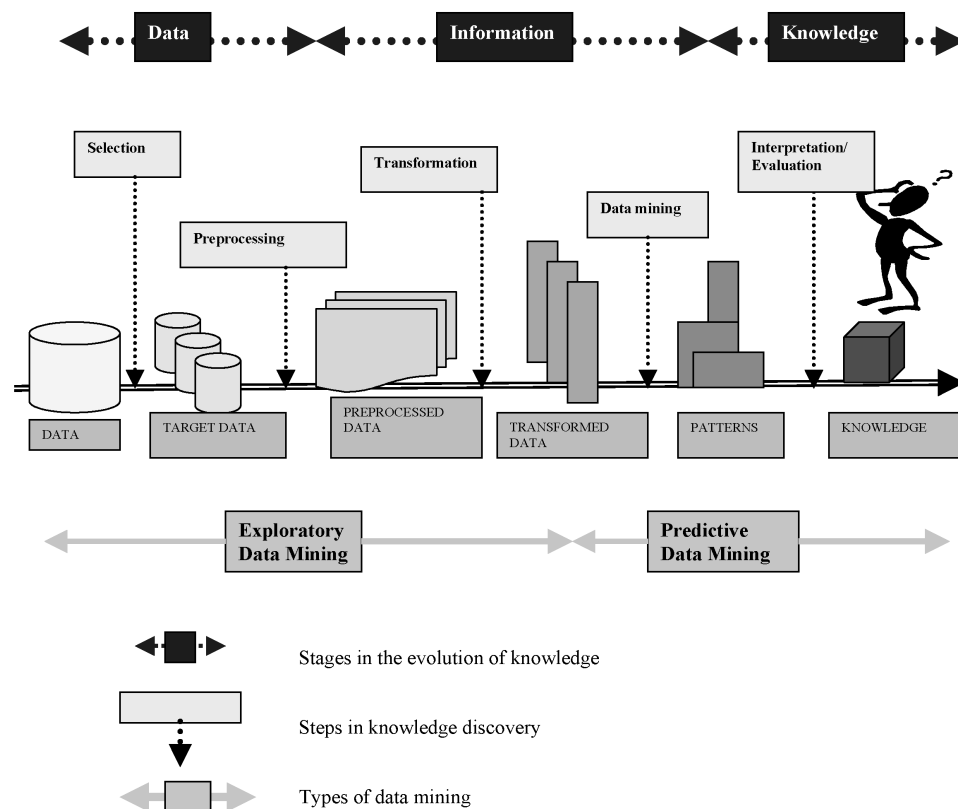
The following steps are typically undertaken in data mining (Chung and Gray, 1999). These steps are iterative, with the process moving backward whenever it is required to do so.

- 1 Develop an understanding of the application, of the relevant prior knowledge and of the end user's goals.
- 2 Create a target data set to be used for discovery.
- 3 Clean and preprocess the data (including handling missing data fields, noise in the data, accounting for time series and known changes).
- 4 Reduce the number of variables and find invariant representations of the data if possible.
- 5 Choose the data mining task (classification, regression, clustering, etc).
- 6 Choose the data mining algorithm.
- 7 Search for patterns of interest (this is the actual data mining).

- 8 Interpret the patterns mined. If necessary, iterate through any of steps 1–7.
- 9 Consolidate the knowledge discovered, prepare reports and then use/reuse the newly created knowledge.

Figure 4 presents a generic representation of a typical knowledge discovery process. A data mining project usually starts with data collection or data selection, covering almost all steps (described above and illustrated in Figure 4) in the KDD process. In this respect, the first three steps of the KDD process (i.e., selection, preprocessing and transformation) are considered to be exploratory data mining, whereas the last two steps (i.e., data mining and interpretation/evaluation) in the KDD process are considered to be predictive data mining. The primary objectives of data mining in practice tend to be description (performed by exploratory data mining) and prediction (performed by predictive data mining). Description focuses on finding human interpretable patterns describing the data while prediction involves using some observations and attributes to predict unknown or future values of other attributes of interest. The relative importance of description and prediction for particular data mining applications can vary considerably. The descriptive and predictive tasks are carried out by applying different machine learning, artificial intelligence and statistical algorithms.

Figure 4 Integrated view of the knowledge discovery process



A major goal of exploratory data mining is data cleaning and understanding. Some of the data operations undertaken during exploratory data mining include: sampling, partitioning, charting, graphing, associating, clustering, transforming, filtering and imputing. Predictive data mining deals with future values of variables and utilises many algorithms; including regression, decision trees and neural networks. Predictive data mining also involves an assessment step which compares the effectiveness of different models according to many performance metrics.

Figure 4 shows an integrated view of the knowledge discovery process, the evolution of knowledge from data to information to knowledge and the types of data mining (exploratory and predictive) and their interrelationships. In this one figure, all the major aspects connected with data mining are captured and by so doing, the integral role of data mining in knowledge creation is emphasised. This is not normally explicitly articulated in the existing literature although the connection between data, information and knowledge is often discussed; for example (Spiegler, 2003).

4 Data mining as an enabler for realising the knowledge spiral: a meta-framework

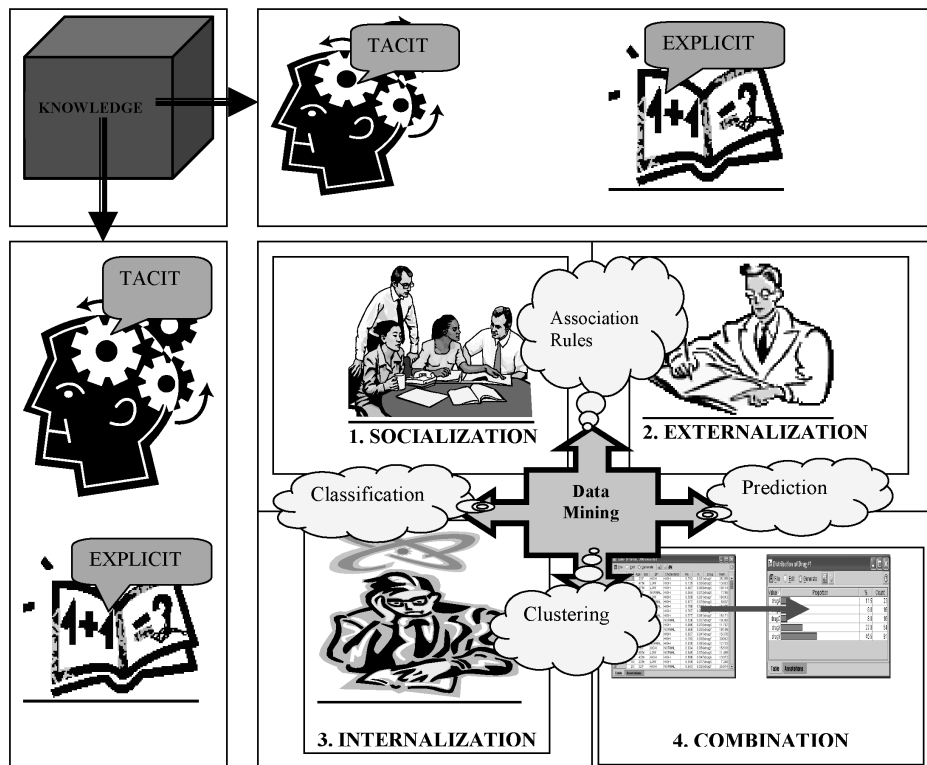
One of the key strengths of data mining as a tool for facilitating knowledge discovery and thereby potentially its benefits to organisations is concerned with its ability to enable the realisation of the knowledge spiral and thus supports both the subjective and objective components of knowledge. In this way, data mining supports the sense making activities as well as the Lokean/Leibnizian functions and thus the information processing activities of knowledge management.

Traditionally, in data mining, the knowledge component depicted in Figure 4 is considered as a homogeneous construct; however as discussed earlier knowledge is indeed a multifaceted construct, reflecting tacit as well as explicit components. This is illustrated in Figure 5 which shows the knowledge component, the types of knowledge and the knowledge transformations of the knowledge spiral and integration of data mining. Specifically, we separate the knowledge component (the end result of data mining) in the KDD process into its constituent parts and we then superimpose the macro and micro processes of knowledge creation/acquisition; namely, Nonaka's knowledge spiral with the specific modes of transformation and the role of data mining in realising these key knowledge creation/acquisition transformations. Whereas Nonaka's knowledge spiral provides macro insights into the types of knowledge and knowledge transformations and thus is clearly beneficial when trying to create and acquire knowledge, data mining enables micro inductive knowledge creation/acquisition from organisational data repositories and knowledge assets.

Data mining outcomes include rules associating attributes with other attributes, prediction, clustering and classification. Since these outcomes are automatically generated, they become more beneficial when coupled with assessments by human experts. Further, when these outcomes are examined by a team of experts, a potentially conducive setting for tacit to tacit transformation (i.e., socialisation) is thus created and hence a forum for narrativising (Boland and Tenkasi, 1995) the knowledge asset (which will be elaborated upon in the discussion section) develops. While socialisation can occur for reasons other than interpreting outcomes of data mining, the structured nature of the outcomes of data mining provides a more systematic basis for guiding such

interactions and thus demonstrating the relationship between data mining and this transformation within the knowledge spiral. In the case of individual interpretations of the outcomes of data mining, the explicit to tacit transformation (internalisation) is induced. Whether expert teams or individual experts are interpreting the outcomes from data mining, they are then more likely to expand these outcomes by externalising some of their own tacit knowledge (i.e., tacit to explicit knowledge transformation is invoked). Finally, operations such as aggregation and visual presentation of data represent explicit to explicit knowledge transformation (i.e., combination). Thus by combining Nonaka's knowledge spiral with data mining we present an integrated meta-framework for knowledge creation which provides a unified view of knowledge creation through joining of the two predominant knowledge creation/acquisition processes. Consequently, we have created a more structured and comprehensive theoretical orientation to addressing the area of knowledge creation/acquisition.

Figure 5 Integrating data mining with the knowledge spiral



Specific Modes of Transformations and the Role of Data Mining in Realizing Them:

1. Socialization – experiential to experiential via practice group interactions on data mining results
2. Externalization – written extensions to data mining results
3. Internalization – enhancing experiential skills by learning from data mining results
4. Combination – gaining insights through applying various data mining visualization techniques

To further elaborate on the capabilities of data mining in this context, let us consider each of the types of knowledge transformations described in Section 2.2. We depict these in Table 1. The key to the power of data mining is that it offers the potential to help actualise all four modes of transformation of the knowledge spiral. Our meta-framework therefore enables better capturing of the dynamics of knowledge creation/acquisition in any organisational setting. The implications of this are far reaching, such as providing a grounded approach to designing systems for knowledge management or enabling a more in depth understanding of knowledge sharing networks in firms.

Table 1 Data mining as an enabler of the knowledge spiral

<i>From</i>	<i>To</i>	
	<i>TACIT</i> ¹	<i>EXPLICIT</i>
<i>TACIT</i>	Interpreting patterns, for example in healthcare diagnosis discovered through data mining enables interaction among physicians – hence making it possible to stimulate and grow their own tacit knowledge.	Interpretation of findings from data mining helps the revealing of tacit knowledge, which can then be articulated and stored as explicit cases in the case repository.
<i>EXPLICIT</i>	By assimilating and internalising knowledge discovered through data mining, experts can turn this explicit knowledge into tacit knowledge which they can apply to new scenarios.	By performing exploratory data mining, such as summarisation and visualisation, it is possible to upgrade, expand and/or revise current facts and protocols.

¹Each entry explains how data mining enables the knowledge transformation from the type of knowledge in the cell's row to the type of knowledge in the cell's column.

5 An illustration of the meta-framework in healthcare

It is possible to illustrate specifically how our meta-framework integrates knowledge creation through data mining and the role of data mining in the realisation of the modes of transformation of the knowledge spiral, using the following healthcare data set (Table 2).² Healthcare a knowledge intensive industry that consists of rich data repositories populated with various patient information and utilisation records and thus is an appropriate choice.

Table 2 Drugs administered to patients

<i>Patient ID</i>	<i>Drug</i>
1	D1, D2
2	D3, D4, D5
3	D3, D1, D2
4	D3, D5, D1
5	D5, D2

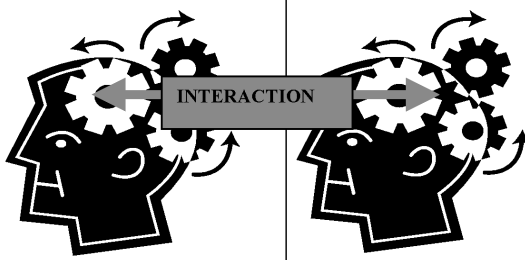
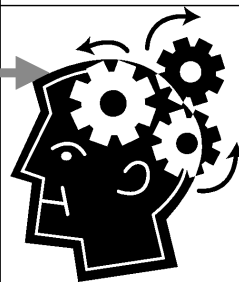
Table 2 is an example of explicit knowledge stored in a medication data repository. Using data mining, the following patterns can be revealed:

- D1 is administered to 60% of the patients (i.e., 3/5)
- D1 and D2 are administered together to 40% of the patients (i.e., 2/5)
- D2 is administered to 67% of the patients who are given drug D1 (i.e., 2/3).

As the physicians try to understand these findings, one physician could explain that D2 has to be given with D1 for patients who had a heart attack at age 40 or less. Thus the following rule can be added to the rule repository: If a patient's age is ≤ 40 years and the patient had a heart attack and D1 is administered to the patient, then D2 should also be administered to that patient. This is an example of existing tacit knowledge (since it originated from the physician's head) transferring to new explicit knowledge that is now recorded for everyone to use.

As physicians discuss the implications of these findings, tacit knowledge from some physicians gets transformed into tacit knowledge for other physicians. Thus, during the interaction of physicians from different specialties, an environment of existing tacit to new tacit knowledge transformation occurs, which consequently realises the sense making component of the Yin-Yang model in Figure 3. Table 3 summarises such knowledge transformations using the data set presented in Table 2.

Table 3 Data mining as an enabler of the knowledge spiral: an example from healthcare

FROM \ TO		TACIT ¹	EXPLICIT
TACIT			If a patient's age is ≤ 40 years and the patient had a heart attack and D1 is administered to the patient, then D2 should also be administered to that patient. (i.e., captured through the a physician's narrating of diagnosis and treatment protocols)
EXPLICIT			• D1 is administered to 60% of the patients. • D1 and D2 are administered together to 40% of the patients. • D2 is administered to 67% of the patients who are given drug D1.

¹Each entry gives an example of how data mining enables the knowledge transformation from the knowledge type in the cell's row to the knowledge type of the cell's column. In this case the predominant data mining technique is association rules

6 Discussion

The primary objective in this paper was to address an important challenge in knowledge management; namely, knowledge creation. In Figure 2 the key dimensions of knowledge management were outlined and creation/acquisition was noted to be not only the first step in the knowledge management process, but critical, since it also serves as the input for the whole knowledge management process. From a systems dynamics point of view, we know that inputs, in particular high quality inputs, are an integral part of any well functioning system, thus underscoring the importance of focusing on knowledge creation. The proposed meta-framework is intended to assimilate the two major mechanisms of knowledge creation into an integrated framework and thereby provide a systematic holistic approach to help and facilitate the creation/generation of the key input for knowledge management viz. knowledge.

Knowledge creation and continuous knowledge renewal grow knowledge and consequently, the value of an organisation's knowledge assets which is essential for better performance in today's knowledge based economy. To date, while there is a wide consensus on the significance of knowledge creation, the approach to its creation has been generally viewed in a somewhat narrow and compartmentalised manner (Schultze and Leidner, 2002; Massey et al., 2002). In particular, the two dominant mechanisms of knowledge creation (Nonaka's knowledge spiral and data mining) so far have been used either separately or independently. A more systematic, synergistic and powerful mechanism for knowledge creation is achieved through combining these two mechanisms and using them interdependently rather than independently. The proposed meta-framework conceptually shows the potential of such an interrelationship and synergistic effect on knowledge creation; in particular how data mining can help the realisation of the knowledge spiral.

Nonaka's knowledge spiral provides a solid theoretical macro foundation for understanding the dynamics of knowledge creation as well as recognising the types of knowledge and the nature of duality that this phenomenon exhibits, as discussed in Section 2.1. A primary limitation with this seminal framework, though, is that it does not provide sufficient detail on how to actualise the specific transformations that it proposes i.e., socialisation, externalisation, internalisation and combination. Thus, it is not discrete enough at the micro level for organisations as they try to embrace knowledge management and develop and design the appropriate strategies, processes, techniques and technologies to support their knowledge management initiatives.

Data mining on the other hand, provides us with a rich tool set of techniques that enables the creation, generation and extraction of knowledge from large amounts of data and information, particularly at the micro level. However, data mining does not seem to reflect the duality of knowledge; i.e., that it is a complex construct, rather it has traditionally treated knowledge as a homogenous 'block' or object that is the end result of the application of the various techniques.

Each of these two approaches to knowledge creation has its particular respective strengths and critical role in facilitating knowledge creation. However, by combining them in the proposed meta-framework, not only is it possible to leverage their respective benefits, it is also possible to achieve the synergy of combining their macro and micro perspectives and also hopefully and at least partially, to compensate for the limitations of each method. Combining key approaches to maximise their respective strengths and at the same time attempting to reduce their respective limitations has been at the core of

IS research and central to the development of sound meta-frameworks. The classic example of such a strategy is that of Gorry and Scott Morton (1971) in which they developed the well recognised Information Systems framework that combined Anthony's categories of managerial activity with Simon's categorisation of decision making.

An important feature for any meta-framework is its robustness with respect to existing, established frameworks and theories. The proposed meta-framework displays this characteristic. For example the people, processes and technology dimensions (or the tripartite knowledge management approach), discussed in Section 2, is a documented approach for understanding knowledge management at the organisational level (Massey et al., 2002). Indeed, this perspective is important for revealing that knowledge management is a multifaceted endeavour, that it is not only technology neither is it only organisational behaviour or only business process reengineering; rather, it is the combination of all these components and will only be successful if viewed as such. The proposed meta-framework provides scope for this tripartite perspective in that all the three components are integral to the application of data mining in realising the transformations of the knowledge spiral.

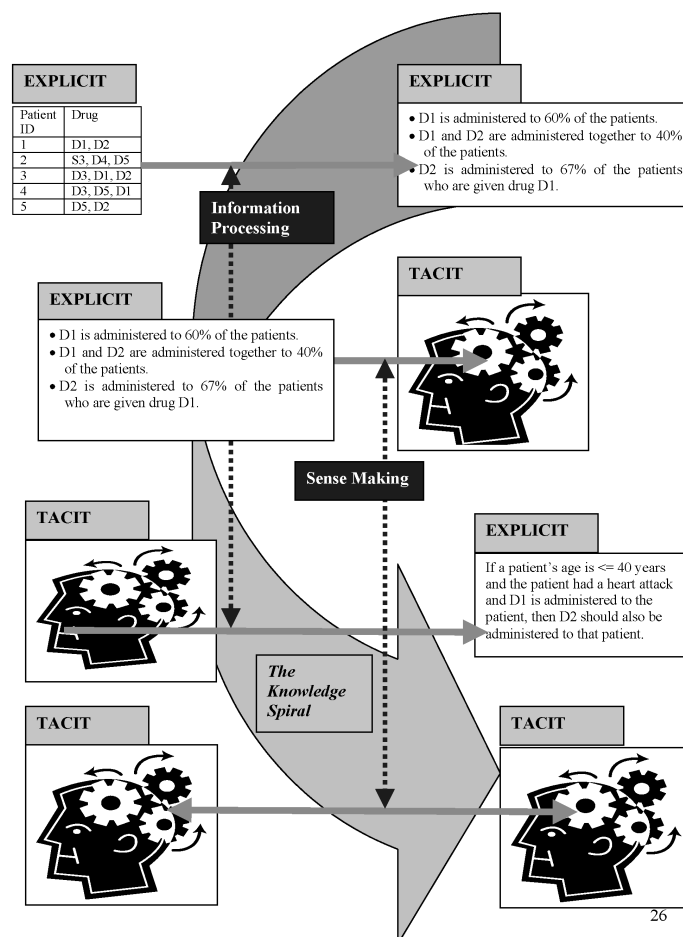
The proposed meta-framework also underscores the duality of knowledge; irrespective of one's ontological preferences with respect to the concept of duality, whether that is Burrell and Morgan's subjective/objective dimensions, Deetz's dimensions of discourse, the Lockean/Leibnizian or the Hegelian/Kantian perspectives identified in the Yin-Yang model of knowledge management. Specifically, Section 2.1 elaborated on this duality by viewing it within the context of the Yin-Yang model, since this model not only presents the duality concept of knowledge but also incorporates the knowledge spiral; both of which are central to the proposed meta-framework.

Knowledge intensive firms are composed of multiple communities of practice with specialised expertise (Boland and Tenkasi, 1995), yet the existence of these communities is not sufficient in itself to lead to knowledge creation. "The narrating of experience is a critically important but often overlooked element of knowledge production ..." (Boland and Tenkasi, 1995), our meta-framework attempts to provide a systematic infrastructure through the outcome from data mining, in the sense of Fleck's concept of the 'thought collective' which emphasises the unique interpretive repertoires of a distinct community of knowing, i.e., the group of experts in the socialisation mode of knowledge transformation. Integral to the changing relationships among communities of knowing via the affecting of perspective making and perspective taking (Boland and Tenkasi, 1995) new knowledge is produced. Successful knowledge management is not only a function of knowledge creation but also and of equal importance, is the interaction of the people vis à vis the created knowledge. The proposed meta-framework creates a structured conduit that links the created knowledge with the communities of knowing and thus strengthens the knowledge domain and practices. In a world that is becoming more and more mobile, knowledge sources then become more diverse and hence the even greater need for a more systematic and all inclusive approach towards knowledge creation/acquisition not only to build knowledge assets but also to help to personalise knowledge dissemination. Furthermore, it is important to note that such a framework could be useful in better understanding knowledge creation/acquisition opportunities within newer information systems such as ERP (enterprise resource planning) systems.

7 Conclusions

Any successful knowledge creation effort should start with a thorough understanding of the problem as well as the data. Exploratory data mining enhances such understanding and possibly enables participants in the data mining process to formulate some hypotheses. The next phase in the knowledge creation process involves a development of some model – association, classification, clustering, or prediction. The model needs to be evaluated – data tested and professionally verified. Where to start in the knowledge creation process is a matter of where an organisation is positioned along a knowledge discovery continuum. At the early stages, one usually starts with a pilot project of a limited scope which serves as a proof of concept for the feasibility and utility of the data mining technology within the given organisational context. Such a pilot requires an organisational championship and a significant amount of data, as the knowledge discovery process is data driven and data intensive by nature. The outcome of the pilot should provide sufficient metrics for evaluating the usefulness of the data mining technology.

Figure 6 Realising the knowledge spiral through data mining



This paper has served to emphasise the key role for data mining in knowledge management by first introducing the need for knowledge management in today's knowledge based economy. Then a discussion of knowledge management, especially identifying the key types of knowledge and the role and benefit of the knowledge spiral was presented. Further, the techniques of data mining and highlighted its role in the knowledge creation process were introduced. Next a meta-framework that integrates data mining within the knowledge spiral and showed how data mining can impact the four modes of transformations of the knowledge spiral was presented. From this, the workings of the proposed meta-framework on a healthcare data set, healthcare being a knowledge intensive industry, was provided. What is of significance is that in today's knowledge based economy, sustainable strategic advantages are gained more from an organisation's knowledge assets than from its more traditional types of assets. Therefore, processes, tools, techniques and frameworks that serve to grow the knowledge assets of an organisation and thereby increase their value are strategic necessities to effectively compete in today's economy. Finally, Figure 6 summarises the meta-framework, its application to knowledge creation and its ability to highlight the fact that knowledge is indeed a complex construct.

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Note

¹Note that for simplicity and purposes of illustration a very small idealised data set is used, naturally healthcare data sets are much more complex and are populated with many more data elements.